Belief networks

Chapter 15.1–2

Outline

◊ Conditional independence

◊ Bayesian networks: syntax and semantics

◊ Exact inference

◊ Approximate inference

Independence

Two random variables A B are (absolutely) ˇindependent iff

$$P(A|B) = P(A)$$

or $P(A, B) = P(A|B)P(B) = P(A)P(B)$

e.g., A and B are two coin tosses

If n Boolean variables are independent, the full joint is

$$\mathbf{P}(X_1, \ldots, X_n) = \prod_i \mathbf{P}(X_i)$$

hence can be specified by just n numbers

Absolute independence is a very strong requirement, seldom m

Conditional independence

Consider the dentist problem with three random variables:

$Toothache$, $Cavity$, $Catch$ (steel probe catches in my to

The full joint distribution has $2^3 - 1 = 7$ independent entries

If I have a cavity, the probability that the probe catches in it do
on whether I have a toothache:

(1) $P(Catch|Toothache, Cavity) = P(Catch|Cavity)$

i.e., $Catch$ is conditionally independent of $Toothache$ given C

The same independence holds if I haven't got a cavity:

(2) $P(Catch|Toothache, \neg Cavity) = P(Catch|\neg Cavit$

Conditional independence contd.

Equivalent statements to (1)

(1a) $P(Toothache|Catch, Cavity) = P(Toothache|Cavity$

(1b) $P(Toothache, Catch|Cavity) = P(Toothache|Cavity$
˘Why??

Full joint distribution can now be written as
$$\mathbf{P}(Toothache, Catch, Cavity) = \mathbf{P}(Toothache, Catch|C$$
$$= \mathbf{P}(Toothache|Cavity)\mathbf{P}(Catch|Cavity)\mathbf{P}(Cavity$$
i.e., $2 + 2 + 1 = 5$ independent numbers (equations 1 and 2

Conditional independence contd.

Equivalent statements to (1)

(1a) $P(Toothache|Catch, Cavity) = P(Toothache|Cavity$

$P(Toothache|Catch, Cavity)$
$\quad = P(Catch|Toothache, Cavity)P(Toothache|Cavity)/P(Catch|C$
$\quad = P(Catch|Cavity)P(Toothache|Cavity)/P(Catch|Cavity)$ (fron
$\quad = P(Toothache|Cavity)$

(1b) $P(Toothache, Catch|Cavity) = P(Toothache|Cavity$
˘W̆hy??

$P(Toothache, Catch|Cavity)$
$\quad = P(Toothache|Catch, Cavity)P(Catch|Cavity)$ (product rule)
$\quad = P(Toothache|Cavity)P(Catch|Cavity)$ (from 1a)

Belief networks

A simple, graphical notation for conditional independence asse
and hence for compact specification of full joint distributions

Syntax:

a set of nodes, one per variable
a directed, acyclic graph (link ≈ "directly influences")
a conditional distribution for each node given its parents:
$\mathbf{P}(X_i | Parents(X_i))$

In the simplest case, conditional distribution represented as
a čonditional probability table (CPT)